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Regularization of Cosmological Correlators: A Global Alternative to Point-Splitting and the Adiabatic Paradigm

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Since the foundational work of Bunch and Davies [Proc. R. Soc. A 360, 117 (1978)], de Sitter space regularization has primarily relied on local point-splitting or perturbative WKB-based adiabatic subtractions. We revisit the exact-solution framework of the Mukhanov-Sasaki equation to propose a global, non-perturbative regularization scheme for cosmological correlator functions.

By leveraging the theory of Bessel functions, we demonstrate that correlators can be regularized holistically through the meromorphic continuation of the Fourier integral. Unlike the local coincidental expansions of point-splitting, our approach utilizes center-of-mass coordinates to isolate and tame polynomial divergences within a momentum-space framework. This method recovers the finite power spectrum for the Bunch-Davies vacuum and extends naturally to non-Bunch-Davies initial conditions, eliminating case-by-case ultraviolet and infrared differentiation.

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Furthermore, we implement a classical variational formalism that departs from traditional gauge-invariant perturbation theory and perturbative ADM approaches. This framework provides a rigorous path for obtaining interaction terms, which we analyze under n-dimensional gauge choices to clarify the apparent super-horizon evolution of curvature perturbations. By providing a scale-independent treatment, this work bridges the gap between foundational de Sitter QFT and modern cosmological perturbation theory.

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Introduction

The topic of regularization of cosmological correlators has been discussed extensively throughout the literature [1–6] in the context of inflation. The foundational framework for such treatments was established in these pages by Bunch and Davies [7], who utilized exact Hankel-function solutions to renormalize the stress-energy tensor via the method of point-splitting. While their work demonstrated the power of using exact mode functions, modern applications have largely shifted toward perturbative WKB-based adiabatic subtractions.

Nevertheless, these contemporary renormalization procedures have shown a persistent dependence on an arbitrary renormalization scale, which usually spoils predictability because of the almost scale-invariant nature of the slow-roll scenario [6]. Furthermore, the reliance on perturbative expansions often necessitates a case-by-case treatment of ultraviolet (UV) and infrared (IR) divergences, departing from the holistic mathematical elegance found in the exact-solution paradigm.

In view of this, a regularization procedure that does not depend on an arbitrary scale of renormalization is proposed. By returning to the exact-solution framework pioneered by Bunch and Davies, but transitioning from a local point-splitting expansion to a global Fourier-space treatment, we demonstrate that divergences can be tamed via meromorphic continuation. This approach utilizes center-of-mass coordinates to isolate polynomial divergences, allowing the finite correlator to emerge as a rigorous mathematical consequence of the integral's analytic structure. This method preserves the exactness of the original 1978 approach while providing a non-perturbative path that is naturally suited for both Bunch-Davies and non-Bunch-Davies vacua. The present paper is organized as follows.

The first section develops a variational formalism that reproduces the results of the Theory of Cosmological Perturbations for scalar modes generated from inflation [8]. This is accomplished by obtaining the Mukhanov-Sasaki (MS) action for the scalar comoving curvature perturbation $\mathcal{R}$, on uniform ϕ hyper-surfaces that stands for linear scalar perturbations in a FRLW background. This formalism is then extended to second order perturbation theory. The logic behind this extension is to obtain an interaction lagrangian density for the scalar field $\mathcal{R}$, as the MS action represents a free-masless field and therefore interactions are not considered.

In the second section a critique to the procedure of renormalization applied to Quantum Field Theory (QFT) in curved space-time is made. From this critique the following investigation is motivated. In addition, for completeness the variational formalism developed previously is applied to derive the standard results of inflationary scalar perturbations and generalize them to n-dimensions. For instance, the MS action is obtained and its equation is solved. The last two subsections are dedicated to connect this variational formalism to the known interaction lagrangian obtained from the perturbative ADM approach of Ref. [9].

In the third section a regularization procedure based on dimensional regularization is presented and applied to calculate the regularized two point correlator of the MS variable in both position and momentum space for a free (non-interacting) theory. This approach constitutes a new *paradigm* rather than a fixed algorithm, in the view that the techniques depend on the model and on the observable to be regularized. Consequently, the framework can be applicable to any QFT model in curved space-time, as well as any QFT in which the In-in formalism is applied to compute correlators. The developed method differs from the common procedures found in the literature of cosmological correlators (Refs. [1,3,4]), in that renormalization by radiative corrections is not needed and thus a renormalization scale is not introduced as a new parameter. Furthermore, the advantage of this formalism is that no counter-term is required because the only counter-terms arising in the exact-solution paradigm come from the point-splitting which we are avoiding —we are not using the DeWitt-Schwinger expansion of equation (2.14) in Ref. [7]. Hence, there is no modification of the underlying theory of gravity. At the end of the section, the known phenomenological result of a quasi-invariant power spectrum is obtained by means of the correlator in momentum space.

1. Variational principle

In inflationary cosmology, the cosmological principle implies a FLRW universe whose metric is

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a^2\gamma_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j \ , \tag{1.1}$$

$$\gamma_{ij} := \delta_{ij} + K\frac{x^i x^j}{1-K\mathbf{x}^2}\,, \qquad i,j=1,2,3\,. \tag{1.2}$$

On the other hand, in order to obtain the Einstein field equations (EinFE) the condition of invariance under
space-time diffeomorphisms, $x \mapsto x + \delta x$, is supposed to be satisfied by the General Relativity (GR) action

$$\mathcal{S}_{\mathrm{GR}} = \int \mathrm{d}^4\,x\,\mathcal{L}_{\mathrm{GR}}\,, \qquad \mathcal{L}_{\mathrm{RG}} := \sqrt{-g}\,\Big(\kappa\,R + L_{\mathrm{m}}\Big)\,, \tag{1.3}$$

where $\kappa = (16\pi G_{\mathrm{N}})^{-1}$, R is the Ricci scalar and L_{m} is the material lagrangian.

In the inflationary context we are interested in a canonical matter lagrangian of the form

$$L_{\mathrm{m}} = -\frac{1}{2}g^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi - V(\phi)\,, \tag{1.4}$$

for a scalar field (in the case of the inflaton $\phi(t)$). The variation of the metric due to the infinitesimal change $x \mapsto x+\delta x$ in the coordinate system is given by

$$\begin{aligned}\varDelta g^{\mu\nu}(x) &:= g^{\mu\nu}(x+\delta x) - g^{\mu\nu}(x)\\ &= \delta g^{\mu\nu}(x) + \delta^2 g^{\mu\nu}(x) + \mathcal{O}\Big(\delta^3 g\Big).\end{aligned} \tag{1.5}$$

Furthermore, from Perturbation Theory applied to General Relativity[1] we know that the first order variation of the metric is equal to its Lie derivative, that we may write as

$$\delta g^{\mu\nu}(x) = \boldsymbol{\nabla}^\mu\xi^\nu + \boldsymbol{\nabla}^\nu\xi^\mu\,, \tag{1.6}$$

where the direction is given by the local change in coordinates, $\xi^\mu(x) := \delta x^\mu$. From this we can notice that if the theory is invariant under an infinitesimal change in coordinates (diffeomorphisms), then we have a gauge freedom in the choice of $\xi^\mu(x)$.

(a) Second order variation

The Scalar-Vector-Tensor decomposition shows that there are only four scalar degrees of freedom (sdf) related to the perturbed metric and the perturbation $\delta\phi(x)$ of the inflaton $\phi(t)$ adds another sdf. Nevertheless, the choice of a gauge diffeomorphism ξ^μ let us discard four sdf, leaving only one sdf with physical significance.

Regarding our approach, we treat one component of the variation ξ^μ as a new scalar field that cannot be eliminated by any gauge choice, allowing us to reproduce a single sdf. This procedure

[1]It is worth to mention that the distinction between the background nonphysical space-time $\mathcal{M}_0$ and the perturbed physical $\mathcal{M}$ may be implicitly made accordingly to the active approach of perturbation theory in general relativity [10]. Nevertheless, in the passive approach we can regard the perturbation of the metric as just a variation in the local frame coordinates, $x \mapsto x+\delta x$, so that the space-time is the same $\mathcal{M}_0 \simeq \mathcal{M}$, and is the metric that suffers the variation according to $\delta g_{\alpha\beta} = \mathcal{L}_\xi g_{\alpha\beta}$.

is proposed as an effective breaking of space-time diffeomorphisms. It suffices to break the time or the spatial invariance, by defining $\varphi := \delta t$ or $\partial^i \varphi := \delta x^i$, to obtain the inflationary dynamics. The new scalar field introduced, for instance $\varphi := \xi^0$, breaks the invariance of the metric under $t \mapsto t + \xi^0$. For that reason it is called a "Goldstone boson" associated with the breaking of time-diffeomorphism (Ref. [2]). However, the symmetry can be restored by supposing that the field transforms as, $\varphi \to \varphi - \zeta^0$, under an arbitrary diffeomorphism: $x^\mu \to x^\mu + \zeta^\mu$.

In the framework of cosmological perturbation theory, selecting this sdf corresponds to the gauge choice

$$\delta g_{\mu\nu} = \begin{pmatrix} -2\dot{\varphi} & -\partial_i \varphi \\ -\partial_i \varphi & 2\,{}^{(0)}g_{ij} H \varphi \end{pmatrix}, \tag{1.7}$$

where ${}^{(0)}g_{ij}$ is the spatial background metric. However, because this mechanism explicitly breaks diffeomorphism invariance, it is not equivalent to the standard treatment which relies on constructing gauge-invariant variables.

To see this one can calculate the gauge-invariant Bardeen potentials defined as

$$\Phi = A - \frac{\mathrm{d}}{\mathrm{d}t}\left(H^{-1} C\right), \tag{1.8a}$$

$$\Psi = B - \dot{E} + 2HE + H^{-1} C\,, \tag{1.8b}$$

for a metric perturbation

$$\delta g_{\mu\nu} = \begin{pmatrix} -2A & \partial_i B \\ \partial_i B & 2\,{}^{(0)}g_{ij} C + 2\partial_i \partial_j E \end{pmatrix}, \tag{1.9}$$

to be $\Phi = \Psi = 0$ for the case where the metric perturbation (1.9) equals (1.7).

This of course makes sense because there cannot be gauge-invariant variables or potentials once the gauge invariance of diffeomorphisms is broken. However, a surprising result is that the MS equation in terms of the induced boson φ can be found, as will be derived in the next section.

Hence, the phenomenological predictions of gauge-invariant perturbation theory remain unchanged at first order, e.g. the primordial power spectrum stays quasi-invariant. This occurs because the metric perturbation (1.7) is effectively gauge-invariant as the non-linear transformation of the bosonic field φ compensates the gauge variation. Consequently, the approach still encodes the single sdf required for calculations, once diffeomorphism symmetry is restored.

In the interest of deriving the MS equation and the interaction terms for the MS variable, we continue with the variations of the action of GR. As mentioned before, following from the variational principle of GR the first background tensor $E_{\mu\nu}$, defined in the appendix (A 3), vanishes at $g^{\alpha\beta}(x)$ for every space-time point and gives the EinFE. Additionally, the Lie derivative $\delta g^{\mu\nu}$ of equation (1.6) shows that the Goldstone boson φ appears with its first derivatives in the first variation of the action as

$$\delta \mathcal{S}_{\mathrm{GR}} = \int \mathrm{d}^4 x \, \frac{\delta \mathcal{L}_{\mathrm{GR}}}{\delta g^{\mu\nu}} \, \delta g^{\mu\nu} = 0\,. \tag{1.10}$$

Hence, because the tensor $E_{\mu\nu}$ vanishes we cannot obtain any dynamical information from (1.10). In spite of that, after doing the second variation of the action, we find that the metric perturbation $\delta g^{\mu\nu}$ appears in quadratic terms

$$\delta^2 \mathcal{S}_{\mathrm{GR}} = \int \mathrm{d}^4 x \, \frac{\delta^2 \mathcal{L}_{\mathrm{GR}}}{\delta g^{\mu\nu} \delta g^{\gamma\lambda}} \, \delta g^{\gamma\lambda} \delta g^{\mu\nu}\,. \tag{1.11}$$

In consequence, this second variation (1.11) will result in the action $\mathcal{S}_\varphi$ of the Goldstone boson φ as the first derivatives of the field appears in quadratic terms. For the computation of the searched action $\mathcal{S}_\varphi$ we begin by writing the variation of second order of the lagrangian in the form

$$\frac{\delta^2 \mathcal{L}_{\rm GR}}{\delta g^{\mu\nu}\delta g^{\gamma\lambda}} = \frac{\sqrt{-g}}{2}\, F_{\gamma\lambda\mu\nu}\ , \tag{1.12}$$

where we have defined the *second background tensor* as[2]

$$\begin{aligned} F_{\gamma\lambda\mu\nu} &:= \frac{\delta E_{\gamma\lambda}}{\delta g^{\mu\nu}} \\ &= g_{\gamma\mu}\, g_{\lambda\nu} \left(\kappa\, R + L_{\rm m}\right) + g_{\gamma\lambda}\left(\frac{1}{2}\partial_\mu\phi\partial_\nu\phi - \kappa\, R_{\mu\nu}\right). \end{aligned} \tag{1.13}$$

In addition, in order to obtain the MS action from the action of φ, the contraction of equation (1.11) ought to be taken to first order in the slow-roll parameter $\varepsilon := -\dot{H}/H^2$ and also has to be associated with a gauge choice that enables the definition of the boson in terms of a metric or material perturbation. This can be obtained by comparing the perturbed metrics, for example $\mathcal{R} = H\varphi$ in the comoving gauge.

The reason for taking only terms $\delta^2\mathcal{S}_{\rm GR}$ of order ε from the active point of view of perturbation theory [10] is to ensure that the perturbation φ belongs to the perturbed quasi-de Sitter space-time and not to the background de Sitter space-time. The difference between this two space-times is that the parameter takes the value $0 < \varepsilon < 1$ and $\varepsilon = 0$, respectively.

On the contrary, from the passive perspective the condition $\varepsilon = 0$ still considers a scalar perturbation φ, but its associated spectrum results completely scale invariant and this does not match inflationary predictions. Finally, it is worth noting that this formalism allow us to neglect frontier terms in the action which are rather quite tedious to work with.

(b) Third order variation

As calculated in Ref. [9] the perturbative ADM formalism gives terms only proportional to $\varepsilon^2\dot{\mathcal{R}}\mathcal{R}^2$ (that are of second order in slow-roll), from which the trispectrum can be computed at tree level.

In our approach this can also be done by considering the terms coming from the variation of third order of the GR action $\delta^3\mathcal{S}_{\rm GR}$, from which the usual bispectrum of single-field inflation can be compute and correct by adding tree level and one-loop interactions. Additionally, is worth mentioning that such an expression for the trispectrum represents a measure of the non-gaussianities arising from auto-interactions [9].

Nevertheless, the quantum corrections to the tree-level, one-loop, and higher-order interactions are left for future work. In this paper we compute only the interaction Lagrangian to complete the variational formalism.

To obtain interactions we calculate the third order variation of the GR action. This variation takes the form

$$\delta^3\mathcal{S}_{\rm GR} = \int \mathrm{d}x^4\ \frac{\delta^3\mathcal{L}_{\rm GR}}{\delta g^{\alpha\beta}\delta g^{\mu\nu}\delta g^{\gamma\lambda}}\ \delta g^{\alpha\beta}\delta g^{\mu\nu}\delta g^{\gamma\lambda}\ , \tag{1.14}$$

where the third order variation of the langragian can be written as

$$\frac{\delta^3\mathcal{L}_{\rm GR}}{\delta g^{\alpha\beta}\delta g^{\mu\nu}\delta g^{\gamma\lambda}} = \frac{\sqrt{-g}}{2}\, H_{\gamma\lambda\mu\nu\alpha\beta}\ , \tag{1.15}$$

[2]Here the computation of the functional derivative in equation (1.13) is being made by neglecting the term $\delta R_{\mu\nu}\delta g^{\mu\nu}$. This term disappear because δ and $\boldsymbol{\nabla}_\sigma$ are interchangeable; where $\delta R_{\mu\nu} = \boldsymbol{\nabla}_\sigma\left(\delta\Gamma^{\sigma}{}_{\mu\nu}\right) - \boldsymbol{\nabla}_\nu\left(\delta\Gamma^{\sigma}{}_{\sigma\mu}\right)$, so that $\delta R_{\mu\nu}\delta g^{\mu\nu}$ sums to a four-divergence and also terms that are proportional to $\boldsymbol{\nabla}_\sigma g^{\mu\nu} = 0$.

with the introduction of the *third background tensor*

$$\begin{aligned} H_{\gamma\lambda\mu\nu\alpha\beta} :=& -\frac{1}{2}\, g_{\alpha\beta}\, F_{\gamma\lambda\mu\nu} + \frac{\delta F_{\gamma\lambda\mu\nu}}{\delta g^{\alpha\beta}}\,, \\ =& -\frac{1}{2}\, g_{\alpha\beta}\, F_{\gamma\lambda\mu\nu} \\ & - \Big(g_{\gamma\alpha} g_{\beta\mu} g_{\lambda\nu} + g_{\gamma\mu} g_{\lambda\alpha} g_{\beta\nu} \Big) \Big(\kappa R + \mathrm{L_m} \Big) \\ & + g_{\gamma\mu} g_{\lambda\nu} \left(\kappa R_{\alpha\beta} - \frac{1}{2} \partial_\alpha \phi \partial_\beta \phi \right) \\ & + g_{\gamma\alpha} g_{\beta\lambda} \left(\kappa R_{\mu\nu} - \frac{1}{2} \partial_\mu \phi \partial_\nu \phi \right). \end{aligned} \tag{1.16}$$

Particularly, in (1.14) the terms of interest are the ones with space-time derivatives of the boson φ. This is due to the fact that the appearance of wave numbers k^μ when calculating the amplitudes in the momentum space implies that the terms of the type: $\delta g^{00} = 2\dot{\varphi}$ and $\delta g^{0i} = -a^{-2}\partial^i\varphi$, represent interactions that are not gauge redundancies. Otherwise, the terms of the type $\delta g^{ij} = -2g^{ij} H\varphi$ can be integrated out —as happens to the mass term of the boson in the MS action.

2. Quantum correlators

(a) Renormalization: regularization via minimal subtraction

As it was stated in the Introduction, the goal of this paper is to calculate equal-time correlators of QFT in curved space-time. A problem that arises in this context is the divergence of correlators. In this work, the procedure of dimensional regularization is chosen, as it is the commonly used in the *renormalization* of QFT, in order to add the so-called *radiative corrections*.

The following discussion on dimensional regularization is based on Refs. [7]& [11].

Nevertheless, in this subsection it is argued that the renormalization may not be the correct paradigm to work with the infinities of a curved space-time background. In the interest of this critique, an explanation of how the preliminary calculations using renormalization would have been implemented is developed.

The regularization in this case would had to be applied to the dimensions of space only, and not to the dimensions of space-time as a whole. This represents a key difference between the In-out and the In-in (also called Schwinger-Keldysh) formalism, and occurs because the integrals of time and space are needed to be calculated separately in this In-in formalism. More explicitly, if there appears a divergent integral, I, when calculating correlators in the momentum space, this divergence is going to be associated with the dimension of space, $n = 3$.

Another important difference between the formalism of renormalization applied to curved space-time, and the one applied to QFT in Minkowski space, resides in that divergences in Minkowski are related to **interactions between fields** but not to **interactions between fields and the space-time** [12]. Of course, this is due to the fact that Minkowski space is non-dynamic. On the other hand, if fields interact with geometry, the renormalization procedure depends on the curvature of the background. Nonetheless, the following discussion applies to both theories, by keeping in mind that the dimension appearing in the calculations has to present a critical behavior.

From the procedure of regularization it follows that the divergent integral needs to be treated as a function of the space (or space-time) dimension, $I = I(n)$. This function is then *analytically continued* by identifying $n \to \omega \in \mathbb{C}$, $I(n) \to I(\omega)$. This integral is expanded near the *pole* where the divergence happens, by taking $\omega = n + \epsilon$. For example, let the Laurent expansion of the integral be

$$I(\omega = n + \epsilon) = a_1(n)\,\epsilon^{-1} + b_0(n) + \mathcal{O}(\epsilon)\,. \tag{2.1}$$

We strongly stress that equation (2.1) is an **approximation** to the real value of the divergent integral, which is valid when the vacuum is well-behaved or adiabatic: it presents almost no vacuum fluctuactions.

The **schema of minimal subtraction** dictates that an infinite quantity, called a **counter-term** $\delta I(n)$ needs to be added to the original lagrangian density in order to cancel the divergent value of such integral $I(\omega)$. Only then the integral is said to be *renormalized*. In the example (2.1), the renormalized integral would be written as

$$I_{\rm ren} := I(n+\epsilon) - \delta I(n), \quad \delta I(n) := a_1(n)\epsilon^{-1} . \tag{2.2}$$

In addition, in Minkowski space-time the infinite quantity $\delta I(n)$ has to be defined in terms of the coupling parameter $\lambda(n)$ of the interaction. This coupling parameter has to depend on the dimension n, in order for the dimensions of energy and space to be consistent. Usually, this dependence of the coupling parameter $\lambda(n)$ on the dimension n is introduced quite artificially by the appearance of the so-called *renormalization scale* μ. Nevertheless, the physical interpretation of this scale is well understood in Minkowski space-time.

More specifically, in a theory with a *bare* coupling constant λ_0, this constant would turn into a *renormalized* coupling constant, $\lambda(\mu)$, by means of the substitution $\lambda_0 = \mu^\epsilon\, Z_\lambda\, \lambda(\mu)$. The relation between the divergence of the integral $I(n+\epsilon)$ and the redefinition of the coupling constant λ_0, by $\lambda(\mu)$, is that the loop corrections actually determine the *Renormalization Group Equation*. From this relation the so-called *beta function*, defined by

$$\beta(\lambda) = \mu \frac{\mathrm{d}\lambda(\mu)}{\mathrm{d}\mu} , \tag{2.3}$$

is calculated. Finally, this beta function will show the explicit dependence of the whole interaction on the scale.

However, in the case of a curved space-time background: 1) the vacuum is usually non-adiabatic, and 2) the counter-terms are dependent on the curvature of the geometry. These facts make the mentioned approach quite ambiguous, as the interactions can be more significant than the bare theory. Furthermore, in principle, the quantum contributions can represent not only small radiative corrections—as those found in the original point-splitting derivations of the Bunch-Davies vacuum (See [7], equation (2.14))—but rather complete redefinitions of the whole theory of gravity.

We see these problems as sufficient reason to compute the exact divergent integral **globally**, by meromorphic continuation, instead of approximating it **locally** by the Laurent expansion and then adding radiative corrections to erase the divergent terms.

(b) Mukhanov-Sasaki in n-dimension

In concordance with the previous critique, we compute the action of the MS field ν and its auto-interaction lagrangian in a flat FLRW $(1+n)$-space-time. This is done in view that our alternative approach of meromorphic continuation utilizes dimensional regularization to regularize these interactions in future work.

In order to compute the interactions, we calculate the contractions of the second (1.13) and third (1.16) background tensors with the metric variations, respectively. Such as it was mentioned in subsections 1(a) and 1(b) the contractions are calculated by means of the EinFE. Additionally, the EinFE in the general case of a $(1+n)$-dimensional space-time background are

$$\kappa R + \mathrm{L_m} = nH^2 - \varepsilon H^2 , \tag{2.4a}$$

$$\kappa R_{\alpha\beta} - \frac{1}{2}\partial_\alpha\phi\partial_\beta\phi = \frac{1}{2} g_{\alpha\beta}\left(nH^2 - \varepsilon H^2\right) . \tag{2.4b}$$

From this equations (2.4), after arranging the terms of first order in slow-roll, the second background tensor has the form

$$F_{\gamma\lambda\mu\nu} = \mathcal{O}\left(\varepsilon^0\right) + \varepsilon H^2\left(\frac{1}{2}g_{\gamma\lambda}g_{\mu\nu} - g_{\gamma\mu}g_{\lambda\nu}\right). \tag{2.5}$$

Then, by using (1.7) and (2.5), the second variation in the GR action (1.11) can be expressed as

$$\begin{aligned}\delta\mathcal{S}_{\mathrm{GR}} = \mathcal{O}&\left(\varepsilon^0\right)\\ &+ \int \mathrm{d}t\, \mathrm{d}^n\, x\, a^n\, \varepsilon H^2\left(-\dot{\varphi}^2 + g^{ij}\partial_i\varphi\partial_j\varphi\right)\\ &+ \mathcal{O}\left(\dot{\varepsilon} + \varepsilon^2\right),\end{aligned} \tag{2.6}$$

after integrating out a mass term for the boson.

The terms of (2.6) have been separated into: a zero order term in slow-roll, which is regarded as a pure gauge redundancy; a first order term that represents the action of a free field in a dynamical background; and higher order terms that vanish in the slow-roll approximation $\varepsilon \approx \text{constant} \ll 1$.

In addition, it is worth to mention that the presence of a mass term in (2.6) is consistent with the fact that the symmetry breaking of a local symmetry gives a massive Goldstone boson. Nevertheless, this apparent mass term vanishes as it can be integrated out into a term of order $\mathcal{O}\left(\varepsilon^0\right)$ in slow-roll [3] .

To connect the above result (2.6) to what is commonly found in the literature (Refs. [1–6,8,9]) a physical quantity has to be defined and thus a gauge has to be fixed. For instance, the comoving gauge where the inflaton ϕ is constant on hyper-surfaces and the comoving perturbation $\mathcal{R}$ is the quantity of interest, can be used. When this is made, the relation

$$\mathcal{R} := H\varphi, \tag{2.7}$$

is identified by means of comparing the metric perturbation of the Goldstone boson (1.7) with the metric perturbation (1.9) in the comoving gauge, where $E = 0$, $C := \mathcal{R}$ and $\delta\phi = 0$.

The action for the curvature perturbation (2.7) can be written as

$$\begin{aligned}\delta\mathcal{S}_{\mathrm{GR}} = \mathcal{O}&\left(\varepsilon^0\right)\\ &+ \int \mathrm{d}\tau\, \mathrm{d}^n\, x\, a^{n-1}\, \varepsilon\left(-\mathcal{R}'^2 + \delta^{ij}\partial_i\mathcal{R}\,\partial_j\mathcal{R}\right)\\ &+ \mathcal{O}\left(\dot{\varepsilon} + \varepsilon^2\right),\end{aligned} \tag{2.8}$$

where a mass term for $\mathcal{R}$ appears in the action (2.8), but is also integrated out. From (2.8) the action for the MS field can be obtained by means of the variables

$$\mathrm{v} := \mathrm{z}\,\mathcal{R}, \quad \mathrm{z} := a^{(n-1)/2}\sqrt{2\varepsilon}\,. \tag{2.9}$$

This action turns out to be

$$\begin{aligned}\delta\mathcal{S}_{\mathrm{GR}} = \mathcal{O}&\left(\varepsilon^0\right)\\ &+ \frac{1}{2}\int \mathrm{d}\tau\, \mathrm{d}^n\, x\left(-\mathrm{v}'^2 + \delta^{ij}\partial_i\mathrm{v}\partial_j\mathrm{v} - \frac{\mathrm{z}''}{\mathrm{z}}\mathrm{v}^2\right)\\ &+ \mathcal{O}\left(\dot{\varepsilon} + \varepsilon^2\right).\end{aligned} \tag{2.10}$$

[3] A rather unusual interpretation of this *effective symmetry breaking* occurs by remembering that the boson was defined as a variation in the time diffeomorphism, $\varphi = \delta t$, it follows that the posterior quantization of this scalar degree of freedom can be interpreted as the *quantization* of time within the scope of a semi-classical theory of gravity.

Thus, in the second line the action of a canonically normalized field has been obtained and can be defined as

$$\mathcal{S}_{\mathrm{MS}} := \frac{1}{2}\int \mathrm{d}\tau\, \mathrm{d}^n x \left(-\mathrm{v}'^2 + \delta^{ij}\partial_i \mathrm{v} \partial_j \mathrm{v} - \frac{\mathrm{z}''}{\mathrm{z}}\mathrm{v}^2 \right), \tag{2.11}$$

for the so-called Mukhanov-Sasaki variable v.

Additionally, the equations of motion for (2.11) can be obtained by means of the variational principle, $\delta\mathcal{S}_{\mathrm{MS}} = 0$. In fact, this gives the MS equation

$$\mathrm{v}'' - \boldsymbol{\nabla}^2 \mathrm{v} - \frac{\mathrm{z}''}{\mathrm{z}}\mathrm{v} = 0\,, \tag{2.12}$$

that in the momentum space[4] can be written as the equation of motion of a harmonic oscillator with a time-dependent frequency,

$$\mathrm{v}_{\boldsymbol{k}}'' + \left(k^2 - \frac{\mathrm{z}''}{\mathrm{z}}\right)\mathrm{v}_{\boldsymbol{k}} = 0\,. \tag{2.14}$$

Furthermore, in the slow-roll approximation the time-dependent frequency can be expressed as

$$\frac{\mathrm{z}''}{\mathrm{z}} = \frac{1}{\tau^2}\left(\nu^2 - \frac{1}{4}\right) + \mathcal{O}\left(\varepsilon^2\right), \quad \nu := \frac{n}{2} + \frac{n+1}{2}\varepsilon - \delta\,. \tag{2.15}$$

It is a standard convention from the literature of inflationary cosmology (Refs. [5,8]), that the solution of (2.14) should be the one that obeys the Bunch-Davies vacuum (Ref. [12]) prescription

$$\mathrm{v}_{\boldsymbol{k}} \to \frac{e^{-\mathrm{i}k\tau}}{\sqrt{2k}}, \quad \tau \to -\infty\,. \tag{2.16}$$

In consequence, the standard result of a Hankel solution (Refs. [5,8]) is generalized to n-spatial dimensions as

$$\mathrm{v}_{\boldsymbol{k}} = \frac{\sqrt{\pi}}{2} e^{\mathrm{i}(\nu+\frac{1}{2})\frac{\pi}{2}} \sqrt{-\tau}\, H_\nu^{(1)}(-k\tau)\,, \tag{2.17}$$

Another common result is that of a quasi-invariant power spectrum for the comoving perturbation $\mathcal{R}$. This can be obtained by taking the canonical quantization of the previous Hankel-type solution (2.17)

$$\hat{\mathrm{v}}_{\boldsymbol{k}} = \int \frac{\mathrm{d}^n \boldsymbol{k}}{(2\pi)^n}\, e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{z}} \left(\hat{\mathbf{a}}_{\boldsymbol{k}}^\dagger \mathrm{v}_{\boldsymbol{k}} + \hat{\mathbf{a}}_{\boldsymbol{k}} \mathrm{v}_{\boldsymbol{k}}^*\right), \tag{2.18}$$

as well as the local expansion of this Hankel solution near zero [5]. The first condition gives the relation

$$\mathcal{P}_{\mathcal{R}}(\boldsymbol{k};\tau_1,\tau_2) = \mathrm{v}_{\boldsymbol{k}}(\tau_1)\mathrm{v}_{\boldsymbol{k}}^*(\tau_2)\,\mathrm{z}^{-1}(\tau_1)\mathrm{z}^{-1}(\tau_2)\,, \tag{2.20}$$

between the power spectrum and the MK modes, whereas the last condition represents the approximation of the solution (2.17) when the scale of perturbations is super-Hubble and near the de Sitter era $-k\tau \approx k/\mathcal{H} \ll 1$. With this two conditions the power spectrum turns out to be approximated by

$$\mathcal{P}_{\mathcal{R}}(\boldsymbol{k}) \approx \frac{H^2}{4\varepsilon}\frac{1}{k^3}\,, \tag{2.21}$$

where the two variables of conformal time coincide and the index in (2.17) is taken to zero order in slow-roll $2\nu \approx 3$.

[4]The Fourier transformation is regarded as

$$\mathrm{v}(\tau,\mathbf{x}) = \int \frac{\mathrm{d}^n\,\boldsymbol{k}}{(2\pi)^n}\,\mathrm{v}_{\boldsymbol{k}}(\tau)\,e^{\mathrm{i}\boldsymbol{k}\cdot\mathbf{x}}\,. \tag{2.13}$$

[5]The first Hankel function can be expand as

$$H_\nu^{(1)}(\rho \ll 1) \sim \frac{-\mathrm{i}}{\pi}\Gamma(\nu)\left(\frac{2}{\rho}\right)^{\nu}. \tag{2.19}$$

To see that the power spectrum (2.21) is quasi-invariant the normalized power spectrum is defined by

$$\Delta^2_{\mathcal{R}}(\boldsymbol{k}) := \frac{k^3}{2\pi^2}\mathcal{P}_{\mathcal{R}}(\boldsymbol{k})\,, \tag{2.22}$$

and evaluated at horizon crossing $k = aH$, to obtain

$$\Delta^2_{\mathcal{R}}(\boldsymbol{k}) = \frac{1}{8\pi^2}\frac{H^2}{\varepsilon}\bigg|_{k=aH}\,. \tag{2.23}$$

Subsequently, from (2.22), the spectral index is defined as

$$n_{\mathrm{s}} - 1 := \frac{\mathrm{d}}{\mathrm{d}\ln k}\ln \Delta^2_{\mathcal{R}}\,. \tag{2.24}$$

Hence, after a straightforward calculation the relation between the quasi-invariant shape of the spectrum and the slow-roll parameters can be written as

$$n_{\mathrm{s}} = 1 - 2\varepsilon - \eta\,, \quad \eta := \frac{1}{\varepsilon}\frac{\mathrm{d}\varepsilon}{\mathrm{d}N}\,. \tag{2.25}$$

(c) Interactions in n-dimension

From the result (2.4), by arranging only the terms of first order in slow-roll, the third background tensor (1.16) can be written as

$$H_{\gamma\lambda\mu\nu\alpha\beta} = \mathcal{O}\left(\epsilon^0\right) + \varepsilon H^2\left(g_{\gamma\alpha}\,g_{\beta\mu}\,g_{\lambda\nu} + g_{\gamma\mu}\,g_{\lambda\alpha}\,g_{\beta\nu} - \frac{1}{2}\,g_{\mu\nu}\,g_{\gamma\alpha}\,g_{\beta\lambda} + \frac{1}{4}\,g_{\alpha\beta}\,g_{\gamma\lambda}\,g_{\mu\nu}\right). \tag{2.26}$$

It is straightforward to calculate the contraction that appears in the third order variation of the GR action (1.14) by arranging the terms of (2.26) that have common factors of $\delta g^{\sigma\rho}$. Thus, the contraction becomes

$$\delta^3\mathcal{S}_{\mathrm{GR}} = \int \mathrm{d}^n x\,\mathrm{d}\tau\, a\frac{\sqrt{-g}}{2}\,H_{\gamma\lambda\mu\nu\alpha\beta}\,\delta g^{\alpha\beta}\,\delta g^{\mu\nu}\,\delta g^{\gamma\lambda}\,,$$

$$= \int \mathrm{d}^n x\,\mathrm{d}\tau\,\frac{a^{n+1}}{2}\bigg(- 14\,H^2\varepsilon\,\dot\varphi^3 \tag{2.27a}$$

$$- 2n\,H^2\varepsilon\,\dot\varphi^2 H\varphi \tag{2.27b}$$

$$+ 10\,H^2\varepsilon\,\dot\varphi\,\partial_i\varphi\partial^i\varphi \tag{2.27c}$$

$$+ 2\left(2n - 3n^2\right) H^2\varepsilon\,\dot\varphi H^2\varphi^2 \tag{2.27d}$$

$$+ 2\,(6-n)\,H^2\varepsilon\,H\varphi\,\partial_i\varphi\partial^i\varphi \tag{2.27e}$$

$$- 2\left(n^3 - 2n^2 + 8n\right) H^2\varepsilon\,H^3\varphi^3\bigg)\,. \tag{2.27f}$$

From the interaction (2.27*b*) one can obtain

$$H^2\varepsilon\dot\varphi^2 H\varphi = \varepsilon\mathcal{R}\dot{\mathcal{R}}^2 + 2\varepsilon^2 H\dot{\mathcal{R}}\mathcal{R}^2 + \varepsilon^3 H^2\mathcal{R}^3\,, \tag{2.28}$$

where the second term is proportional to the leading term found in Ref. [9]. However, the interaction (2.27*d*) is given by

$$H^2\varepsilon\dot\varphi H^2\varphi^2 = \varepsilon H\dot{\mathcal{R}}\mathcal{R}^2 + \varepsilon^2 H^2\mathcal{R}^3\,, \tag{2.29}$$

so the first term is of first order in slow-roll. The arising of the same contribution term $\dot{\mathcal{R}}\mathcal{R}^2$ with different order in the perturbative expansion of slow-roll represents an ambiguity in our gauge. This ambiguity can also be seen from the fact that the term (2.27*f*) induces an auto-interaction

$\varepsilon\mathcal{R}^3$ that could spoil the conservation of the comoving perturbation outside the horizon. To solve this issue the following criteria is taken:

The order of a contribution term in the effective action is regarded as a gauge choice.

This criteria is analogous to the procedure developed in Ref. [9], where different gauge choices in the perturbed metric gave rise to different fields: one that presented horizon conservation and one that show the leading order term. It is worth to mention that this gauge redundancies appear because the action has been taken to second order in the perturbative expansion, and so now the field definition (2.7) of $\mathcal{R}$ may not represent a quantity that is conserved outside the horizon, as this gauge choice was made to first order in the perturbative expansion.

For instance, in order to make the MS action to be of second order in the perturbative expansion, we define the comoving gauge by the choice of the table 1. In this fashion, the mass

Table 1. *Second order* comoving gauge showing *horizon conservation*.

Gauge choice:	$\varepsilon = \mathcal{O}(1)$, $\partial_\mu \mathcal{R} = \mathcal{O}\left(\frac{1}{2}\right)$, $\mathcal{R} = \mathcal{O}(-1)$.
Dynamical terms	Mass/Auto-interaction terms
(MS) $\varepsilon\dot{\mathcal{R}}^2 = \mathcal{O}(2)$	(MS) $\varepsilon\mathcal{R}^2 = \mathcal{O}(-1)$
$\varepsilon\dot{\mathcal{R}}\mathcal{R}^2 = \mathcal{O}\left(-\frac{1}{2}\right)$	$\varepsilon^2\mathcal{R}^3 = \mathcal{O}(-1)$
$\varepsilon^2\dot{\mathcal{R}}\mathcal{R}^2 = \mathcal{O}\left(\frac{1}{2}\right)$	$\varepsilon^3\mathcal{R}^3 = \mathcal{O}(0)$, $\varepsilon^4\mathcal{R}^3 = \mathcal{O}(1)$

term of the MS action (2.8) is negligible of order $\mathcal{O}(-1)$ —meaning that the term corresponds to a gauge redundancy, so it is justified that this term was integrated out. Furthermore, the cubic auto-interaction terms are smaller than the MS action, thus preserving conservation outside the horizon.

Table 2. *Second order* comoving gauge showing the *leading term*.

Gauge choice:	$\varepsilon = \mathcal{O}(1)$, $\partial_\mu \mathcal{R} = \mathcal{O}\left(\frac{1}{2}\right)$, $\mathcal{R} = \mathcal{O}(0)$.
Dynamical terms	Mass/Auto-interaction terms
(MS) $\varepsilon\dot{\mathcal{R}}^2 = \mathcal{O}(2)$	(MS) $\varepsilon\mathcal{R}^2 = \mathcal{O}(1)$
$\varepsilon\dot{\mathcal{R}}\mathcal{R}^2 = \mathcal{O}\left(1+\frac{1}{2}\right)$	$\varepsilon^2\mathcal{R}^3 = \mathcal{O}(2)$
$\varepsilon^2\dot{\mathcal{R}}\mathcal{R}^2 = \mathcal{O}\left(2+\frac{1}{2}\right)$	$\varepsilon^3\mathcal{R}^3 = \mathcal{O}(3)$, $\varepsilon^4\mathcal{R}^3 = \mathcal{O}(4)$

On the other hand, the table 2 shows that the leading term is $\varepsilon^2\dot{\mathcal{R}}\mathcal{R}^2$, such that this term is bigger than the MK terms by a factor of one half, $\mathcal{O}(2) \lesssim \mathcal{O}\left(2+\frac{1}{2}\right)$. However, in this gauge choice now appears an auto-interaction term, $\varepsilon^2\mathcal{R}^3$, that spoils the horizon conservation to second order in slow-roll. Regardless, as in the slow-roll limit $\varepsilon \ll 1$ this term can be neglected, it is justified that the MK action (2.8) did not consider it.

After this careful examination, it can be concluded that the criteria does indeed resolve the gauge ambiguity. In addition, one can observe that this choice of hierarchy in the perturbative expansion is analogous to the field redefinition and integration of frontier terms found in Ref. [9].

(d) Quantum interactions

Even though it may not be physical, let us examine the cubic interaction (2.27*f*). The interaction hamiltonian density [6] is

$$\mathcal{H}_{\text{int}} = \lambda\, \hat{\mathrm{v}}^3\,, \qquad \lambda(n,\tau) := (n^3 - 2n^2 + 8n)\, H^2\, a^{n+1}\, \varepsilon\, \mathrm{z}^{-3}\,. \tag{2.30}$$

The cubic interaction (2.30) along with the observable $W(\tau,\mathbf{z},\mathbf{z}') = \hat{\mathrm{v}}(\tau,\mathbf{z})\hat{\mathrm{v}}(\tau,\mathbf{z}')$ can be written as a sum of contractions of operators that is only non-vanishing for the one-loop interaction (A 1*c*) (once the expectation value of the vacuum is taken). This is because Wick's theorem is applicable for operators in the interaction picture, and so the products of operators with odd multiplicity vanishes.

With the definition of the variables: $x := (\tau_1, \mathbf{x})$, $y := (\tau_2, \mathbf{y})$, $z := (\tau_\star, \mathbf{z})$ and $z' := (\tau_\star, \mathbf{z}')$, the integrand inside the first term of (A 1*c*) is calculated as

$$\begin{aligned}
\left\langle \mathcal{H}_{\text{int}}(y)\, \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(z')\, \mathcal{H}_{\text{int}}(x) \right\rangle = & \\
\lambda^2 \Big(& 9\, \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(z')\hat{\mathrm{v}}(x)\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \\
& + 18\, \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(z')\hat{\mathrm{v}}(x)\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(x)\rangle^2 \\
& + 18\, \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(x)\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \\
& + 9\, \langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(x)\rangle \left\langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \\
& + 18\, \langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(x)\rangle \left\langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(x)\rangle^2 \\
& + 18\, \langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(x)\rangle \left\langle \hat{\mathrm{v}}(z')\hat{\mathrm{v}}(x)\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(x)\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \\
& + 3\, \left\langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(x)\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \\
& + 6\, \left\langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(x)\rangle^3 \\
& + 6\, \left\langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(x)\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \Big)\,,
\end{aligned} \tag{2.31}$$

whereas the integrand inside the second term becomes

$$\begin{aligned}
\left\langle \mathcal{H}_{\text{int}}(x)\, \mathcal{H}_{\text{int}}(y)\, \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(z') \right\rangle = & \\
\lambda^2 \Big(& 9\, \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \\
& + 18\, \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \\
& + 18\, \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle^2 \\
& + 9\, \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \\
& + 18\, \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \\
& + 18\, \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(z)\rangle \left\langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle^2 \\
& + 3\, \left\langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \\
& + 6\, \left\langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle^3 \\
& + 6\, \left\langle \hat{\mathrm{v}}(z)\hat{\mathrm{v}}(z')\right\rangle \langle \hat{\mathrm{v}}(y)\hat{\mathrm{v}}(y)\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(x)\rangle \Big)\,.
\end{aligned} \tag{2.32}$$

From this results is clear that the two-point correlator or auto-interaction propagator $\langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle$ has to be calculated. In fact, this correlator has to be regularized.

[6] It is worth mentioning that the relation between the hamiltonian and the hamiltonian density is just simply $H_{\text{int}}(\tau) = \int \mathrm{d}^n\mathbf{x}\, \mathcal{H}_{\text{int}}(\tau,\mathbf{x})$, as it is used in the calculations.

3. Regularization via the Theory of Bessel Functions

Because of the ambiguities that arise from the schema of minimal subtraction, an alternative approach is chosen to avoid the addition of curvature-dependent counter-terms, as the higher-order terms can bring inconsistencies with the slow-roll approximation. Instead, a special type of integral that stems from the Theory of Bessel functions (Ref. [13]) can be used to regularize the correlator via an analytic continuation of the space dimension. Of course, this cannot be implemented without a cost.

Let us discuss the steps added in this new procedure:

(i) First, an alternative solution to the Mukhanov-Sasaki equation is needed, as there appears to be only global results for integrals of the Bessel function and not for other solutions of the Bessel equation. For instance, Bunch & Davies (see eq. (2.14) in [7]) used a local result for Hankel functions, by means of the DeWitt-Schwinger expansion.
Even though this solutions (Hankel or Bessel) are mathematically equivalent they vary in physical meaning, so altering the Bunch-Davies vacua may obscure our physical understanding of the problem.

(ii) Second, in order to solve the integral we need to analytically continue the norm of the position vectors. Although unusual, this does not compromise our physical intuition, since analytically continuing physical quantities is a standard technique for handling divergent integrals. Of course, consistency and physical interpretation are key, which are preserved by the functional form of the observable.

Basically, only the first step may imply a cost of physical understanding. As it was mentioned in section 2(b), the standard mode solution to the MK equation is in terms of the first Hankel function (2.17). This solution represents outgoing waves with positive frequency that resemble a Minkowski vacuum in the far past, the so-called "in-region" of space-time. Furthermore, the mode solution gives a "preferable" vacuum for all inertial observers as it is free of particles (Ref. [12]).

On the other hand, by taking the Bessel function as the solution, we are mixing both outgoing and incoming waves with positive and negative frequency modes, respectively, which do not resemble a canonical Minkowski vacuum. This non-canonical vacuum can contain detectable particles by means of observable momenta excitations. In other words, this alternative mode solution can represent a non-empty vacuum (Ref. [12]). This issue is of fundamental-physics nature and should be addressed in future work. At this stage we will focus solely on phenomenological and mathematical conclusions by calculating corrections to the power spectrum. This paradigm implicitly assumes that the results should be independent of the choice of vacua.

Let us begin by taking the alternative solution of

$$\mathrm{v}_{\boldsymbol{k}}(\tau) = \frac{\sqrt{\pi}}{2}\sqrt{-\tau}\, J_\nu(-k\tau)\,, \quad \nu = \frac{n}{2} + \frac{n+1}{2}\,\varepsilon - \delta\,, \tag{3.1}$$

to the MK equation (2.14), where the constant of normalization has been chosen accordingly to the asymptotic expansion of the Bessel function

$$J_\nu(\rho) \to \sqrt{\frac{2}{\pi\rho}}\cos\left(\rho - \nu\frac{\pi}{2} - \frac{\pi}{4}\right), \quad \rho \to \infty\,, \tag{3.2}$$

along with the vacuum prescription of

$$\mathrm{v}_{\boldsymbol{k}} \to \frac{1}{\sqrt{2k}}\cos\left(-k\tau\right), \quad \tau \to -\infty\,, \tag{3.3}$$

which, of course, is different from (2.16)[7] .

[7] Also, by noticing that

$$\cos\left(-k\tau\right) = \frac{e^{\mathrm{i}k\tau} + e^{-\mathrm{i}k\tau}}{2}\,, \tag{3.4}$$

With this new solution the correlator turns out to be

$$\begin{aligned}\langle\hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle &= \int \frac{\mathrm{d}^n\boldsymbol{k}}{(2\pi)^n}\, e^{\mathrm{i}\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{y})}\, \mathrm{v}_{\boldsymbol{k}}(\tau_1)\mathrm{v}^*_{\boldsymbol{k}}(\tau_2)\,,\\ &\approx \Omega_n\, \frac{\pi\sqrt{\tau_1\tau_2}}{4(2\pi)^n}\int_0^\infty \mathrm{d}k\, k^{n-1}\, e^{\mathrm{i}kr}\, J_\nu(-k\tau_1)\, J_\nu(-k\tau_2)\,,\\ r &:= |\mathbf{x}-\mathbf{y}|\,,\end{aligned}$$

where an approximation has been applied to neglect the direction of the momentum vectors.

This integral (3.5*a*) can be calculated by using an old result (Ref. [13]) from the Theory of Bessel functions by Beltrami, Sommerfeld, Gengenbauer and Macdonald (BSGM),

$$\int_0^\infty \mathrm{d}k\, k^{\mu-1}\, e^{-ak}\, J_\nu(bk)\, J_\nu(ck) = \frac{(bc)^\nu\, \Gamma(\mu+2\nu)}{\pi\, a^{\mu+2\nu}\, \Gamma(2\nu+1)}\, I(\mu,\nu,z)\,, \tag{3.5a}$$

$$I(\mu,\nu,z) := \int_0^\pi {}_2F_1\left(\frac{\mu+2\nu}{2}, \frac{\mu+2\nu+1}{2}; \nu+1; z\right)\, \sin^{2\nu}(\phi)\, \mathrm{d}\phi\ , \tag{3.5b}$$

$$z(\phi) := -\frac{b^2}{a^2} - \frac{c^2}{a^2} + 2\frac{bc}{a^2}\, \cos(\phi)\,, \tag{3.5c}$$

$$\Re(a\pm \mathrm{i}b \pm \mathrm{i}c) > 0\,, \quad \Re(\mu+2\nu) > 0\,.$$

By comparison between (3.5*a*) and (3.5) it can be noticed that to fulfilled the restriction of the domain[8] an analytical continuation must be performed in one of the variables: r, τ_1 or τ_2, in order to have a non-vanishing real part[9] $\Re(a \pm \mathrm{i}b \pm \mathrm{i}c) > 0$, where we have defined $a := -\mathrm{i}r$, $b := -\tau_1$, $c := -\tau_2$.

Let us map r to the complex plane by making $r \to r + \mathrm{i}\delta r$, then we have that

$$\Re(a \pm \mathrm{i}b \pm \mathrm{i}c) = \delta r > 0\,,$$

is satisfied, provided that $\delta r > 0$.

In the particular case of slow-roll the last integral (3.5*b*) in the BSGM result can be approximated by inverting the relation (2.15) of the frequency in terms of the space dimension as

$$\begin{aligned} n &= \frac{2(\nu+\delta)-\varepsilon}{1+\varepsilon}\\ &= 2\nu + 2\delta - \varepsilon(1+2\nu) + \mathcal{O}\left(\varepsilon^2, \delta\varepsilon\right)\ ,\\ &\approx 2\nu\,, \end{aligned} \tag{3.6}$$

where the approximation is taken to order zero in slow-roll. In this way the slow-roll parameters, which depend on the curvature of space-time, are not analytically continue. By using $\sin^{2\nu}(\phi) \approx \sin^3(\phi)$, when $2\nu \approx n$ and $n \to 3$, we obtain

$$I := \int_0^\pi f(\mathbf{a}, z(\phi))\, \sin^3(\phi)\, \mathrm{d}\phi\ , \tag{3.7}$$

is clear that we are mixing incoming with outgoing waves, or equivalently negative with positive frequency modes.

[8]The restriction $\Re(\mu) > -\frac{1}{2}$ is trivial because μ , ν are always positive.

[9]In this case, the four equations are the same because b and c are real, but in general one would have to verify the four equations individually.

where we have defined the function

$$f(\mathbf{a}, z) := {}_2F_1\left(a_1, a_2; a_3; z\right), \tag{3.8a}$$

$$\mathbf{a} := (a_1, a_2, a_3), \tag{3.8b}$$

$$a_1 := \frac{\mu + 2\nu}{2}, \tag{3.8c}$$

$$a_2 := \frac{\mu + 2\nu + 1}{2}, \tag{3.8d}$$

$$a_3 := \nu + 1. \tag{3.8e}$$

Also, by using the notation

$$\mathbf{n} := (n, n, n), \quad \ell(\mathbf{a}) := \frac{a_1 a_2}{a_3}, \tag{3.9}$$

the following property of the hypergeometric function

$$\frac{d}{dz}\, {}_2F_1(a, b; c; z) = \frac{ab}{c}\, {}_2F_1(a+1, b+1; c+1; z), \tag{3.10}$$

can be written as

$$f(\mathbf{a}, z) = \frac{1}{\ell(\mathbf{a}-\mathbf{1})} \frac{\mathrm{d}}{\mathrm{d}z} f(\mathbf{a}-\mathbf{1}, z). \tag{3.11}$$

With this property (3.11), the integral (3.7) can be solved by parts with the change of variable introduced in (3.5c). To see this, let us define an auxiliary function in terms of z as

$$u(z) := 1 + A(z+B)^2,$$

$$A := -\frac{a^4}{4b^2\, c^2},$$

$$B := \frac{b^2 + c^2}{a^2},$$

$$\mathrm{d}z = -\frac{1}{\sqrt{-A}} \sin(\phi)\, \mathrm{d}\phi,$$

to obtain

$$I = -\sqrt{-A} \left(u(z) \frac{f(\mathbf{a}-\mathbf{1}, z)}{\ell(\mathbf{a}-\mathbf{1})} - \frac{du(z)}{dz} \frac{f(\mathbf{a}-\mathbf{2}, z)}{\ell(\mathbf{a}-\mathbf{1})\, \ell(\mathbf{a}-\mathbf{2})} + \frac{d^2u(z)}{dz^2} \frac{f(\mathbf{a}-\mathbf{3}, z)}{\ell(\mathbf{a}-\mathbf{1})\, \ell(\mathbf{a}-\mathbf{2})\, \ell(\mathbf{a}-\mathbf{3})} \right) \Bigg|_{z(0)}^{z(\pi)}. \tag{3.12}$$

Consequently, by inserting the result (3.12) into the BSGM integral (3.5), the correlator can be approximated as

$$
\begin{aligned}
\langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle \approx \frac{\Omega_n}{8(2\pi)^n}\,\frac{\Gamma(\mu+2\nu)}{\Gamma(2\nu+1)}
&\left\{ \frac{(\nu)_2\,(\tau_1\tau_2)^{\nu-\frac{3}{2}}}{(\frac{\mu+2\nu-2}{2})_2\,(\frac{\mu+2\nu-1}{2})_2}\,(-\mathrm{i}r+\delta r)^{4-\mu-2\nu} \right.\\
&\left[{}_2F_1\left(\frac{\mu+2\nu-4}{2},\frac{\mu+2\nu-3}{2};\nu-1;-\frac{(\tau_1+\tau_2)^2}{(-\mathrm{i}r+\delta r)^2}\right)\right.\\
&\qquad\left. +\,{}_2F_1\left(\frac{\mu+2\nu-4}{2},\frac{\mu+2\nu-3}{2};\nu-1;-\frac{(\tau_1-\tau_2)^2}{(-\mathrm{i}r+\delta r)^2}\right)\right]\\
&+\frac{(\nu)_3\,(\tau_1\tau_2)^{\nu-\frac{5}{2}}}{2\,(\frac{\mu+2\nu-2}{2})_3\,(\frac{\mu+2\nu-1}{2})_3}\,(-\mathrm{i}r+\delta r)^{6-\mu-2\nu}\\
&\left[{}_2F_1\left(\frac{\mu+2\nu-6}{2},\frac{\mu+2\nu-5}{2};\nu-2;-\frac{(\tau_1+\tau_2)^2}{(-\mathrm{i}r+\delta r)^2}\right)\right.\\
&\qquad\left.\left. -\,{}_2F_1\left(\frac{\mu+2\nu-6}{2},\frac{\mu+2\nu-5}{2};\nu-2;-\frac{(\tau_1-\tau_2)^2}{(-\mathrm{i}r+\delta r)^2}\right)\right]\right\},
\end{aligned}
\tag{3.13}
$$

when $\sin^{2\nu}(\phi) \approx \sin^3(\phi)$ and where $(\alpha)_\ell := \alpha(\alpha-1)\cdots(\alpha-\ell+1)$ is the descending factorial symbol.

At this point, the expression (3.13) has a similar form to the seminal result of Bunch & Davies, equation (2.11) in Ref. [7], 1978. This represents a proof of consistency within our method and previous results, where the difference is expected due to their canonical choice of Hankel functions as vacua.

According to (3.13), we can define the regularized correlator as

$$
\langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle_{\mathrm{reg}} := \sum_{\ell,\,\mathfrak{s}=1}^{2} e_{(\ell,\,\mathfrak{s})}\,A^{(\ell)}\,(-\mathrm{i}r)^{\alpha(\ell)}\,f\left(\mathbf{v}^{(\ell)};c^{(\mathfrak{s})}\,r^{-2}\right),
\tag{3.14a}
$$

$$
e_{(\ell,\,\mathfrak{s})} := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \ell = 1,2\,, \quad \mathfrak{s} = 1,2\,,
\tag{3.14b}
$$

$$
A^{(\ell)} := \frac{\Omega_n}{8(2\pi)^n}\,\frac{\Gamma(\mu+2\nu)}{\Gamma(2\nu+1)}\,\frac{(\nu)_{\ell+1}}{2^{\ell-1}\,(\frac{\mu+2\nu-2}{2})_{\ell+1}\,(\frac{\mu+2\nu-1}{2})_{\ell+1}}\,(\tau_1\tau_2)^{\nu-\frac{1}{2}-\ell}, \quad \ell = 1,2\,,
\tag{3.14c}
$$

$$
\alpha(\ell) := 2+2\ell-\mu-2\nu\,, \quad \ell = 1,2\,,
\tag{3.14d}
$$

$$
\mathbf{v}^{(\ell)} := \left(\frac{\mu+2\nu-2-2\ell}{2},\frac{\mu+2\nu-1-2\ell}{2},\nu-\ell\right), \quad \ell = 1,2\,,
\tag{3.14e}
$$

$$
c^{(\mathfrak{s})} := \left(\tau_1\,(-)^{\mathfrak{s}+1}\,\tau_2\right)^2, \quad \mathfrak{s} = 1,2\,,
\tag{3.14f}
$$

by taking the limit $\delta r \to 0$. For the following calculations, the function (3.14) should be evaluated at some equal conformal time for both $\tau_1 = \tau$, $\tau_2 = \tau$, that will then correspond to the horizon exit of perturbations, $\tau = \tau_\star$. When taking the approximations

$$\mathbf{v}^{(1)} = \left(\frac{\mu + 2\nu - 4}{2}, \frac{\mu + 2\nu - 3}{2}, \nu - 1\right),$$

$$\approx \left(n-2,\, n - \frac{3}{2},\, \frac{n}{2} - 1\right) \to \left(1,\, \frac{3}{2},\, \frac{1}{2}\right), \tag{3.15a}$$

$$\mathbf{v}^{(2)} = \left(\frac{\mu + 2\nu - 6}{2}, \frac{\mu + 2\nu - 5}{2}, \nu - 2\right),$$

$$\approx \left(n-3,\, n - \frac{5}{2},\, \frac{n}{2} - 2\right) \to \left(0,\, \frac{1}{2},\, -\frac{1}{2}\right), \tag{3.15b}$$

as $n \to 3$, we get

$$\langle \hat{\mathrm{v}}(\mathbf{x},\tau)\hat{\mathrm{v}}(\mathbf{y},\tau)\rangle_{\mathrm{reg}} = A^{(1)}\,(-\mathrm{i}r)^{\alpha(1)} + \frac{A^{(1)}}{2}\,(-\mathrm{i})^{\alpha(1)}\, r^{\alpha(1)+2}\left[\frac{1}{(r+\omega)^2} + \frac{1}{(r-\omega)^2}\right], \tag{3.16}$$

$$\omega := |2\tau|\,, \tau \leq 0\,.$$

This result gives a regularized correlator. However, the correlator inherits a regular pole of order two at $r = \omega$ from the analytical continuation (Ref. [14]) of the hypergeometric function ${}_1F_2\left(1, \frac{3}{2}, \frac{1}{2}; (2\tau)^2\, r^{-2}\right)$ outside of the radius of convergence of the gaussian hypergeometric series[10]. The advantage of this continuation is that it does not have a pole at infinity, this will turn out helpful for the calculations in momentum space.

(a) Regularized power spectrum

Now that the finite, regularized equal-time correlator (3.16) has been obtained in position space, the criteria adopted in this work require us to return to momentum space in order to compare the result with the power spectrum commonly used in the literature on cosmological inflation (Refs. [5,8]).

This criteria, as explained in subsection 2(a), gives the following reasoning:

Once we have integrated the correlator along the whole range of momentum and, in consequence, found a regularized result that does not depend on a counter-term (and thus does not depend on the space-time curvature), we can assure that this is the correct regularized correlator. However, because this correlator is in the position representation, the last step ought to be to express this correlator in the momentum representation, in order to connect the result with the power spectrum.

The regularized correlator in the position representation is regarded, using the projectors $|\boldsymbol{k}\rangle\langle\boldsymbol{k}|\,,\ |\boldsymbol{k}'\rangle\langle\boldsymbol{k}'|$, as

$$\langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle_{\mathrm{reg}} = \int \frac{\mathrm{d}^n\boldsymbol{k}}{(2\pi)^n}\frac{\mathrm{d}^n\boldsymbol{k}'}{(2\pi)^n}\, e^{\mathrm{i}(\boldsymbol{k}\cdot\mathbf{x} - \boldsymbol{k}'\mathbf{y})}\, \langle \hat{\mathrm{v}}_{\boldsymbol{k}}(\tau_1)\hat{\mathrm{v}}_{\boldsymbol{k}'}(\tau_2)\rangle_{\mathrm{reg}}\,. \tag{3.17}$$

Then, by identifying

$$\left\langle \hat{\mathrm{v}}(\boldsymbol{k},\tau_1)\hat{\mathrm{v}}(\boldsymbol{k}',\tau_2)\right\rangle_{\mathrm{reg}} = (2\pi)^n\,\mathcal{P}_{\mathrm{v}}(\boldsymbol{k};\tau_1,\tau_2)\,\delta(\boldsymbol{k}-\boldsymbol{k}')\,, \tag{3.18}$$

the regularized power spectrum can be found by means of the inverse Fourier transform of (3.17).

Nevertheless, by using the projectors $|\boldsymbol{x}\rangle\langle\boldsymbol{x}|\,,\ |\boldsymbol{y}\rangle\langle\boldsymbol{y}|$ to obtain the momentum representation, the integration will be written in terms of two variables. Instead, a simpler approach is to exploit the symmetries of the correlator by means of the projectors $|\mathbf{r}\rangle\langle\mathbf{r}|\,,\ |\boldsymbol{\xi}\rangle\langle\boldsymbol{\xi}|$. Such new variables correspond to a relative position type vector $\mathbf{r}$ and a center of mass type vector $\boldsymbol{\xi}$. Then the

[10] The gaussian hypergeometric series has a radius of convergence of $|z| < 1$ for every $z \in \mathbb{C}$. Nonetheless, its analytical continuation can have regular poles at $z = 1$ or at infinity.

integration needs to be performed in one variable as the correlator (3.14) only depends on r. This approach or approximation is the same as the applied to simplify the integration of (3.13) when using $r = |\mathbf{x} - \mathbf{y}|$ in (3.5*a*): we are assuming that the correlator satisfies the necessary symmetries for the $2n$ degrees of freedom of the two variables $\mathbf{x}$ and $\mathbf{y}$ to be simplified to just n degrees of the variable $\mathbf{r}$, also by eventually taking the integration over r we are assuming isotropy and thus neglecting $n-1$ degrees.

Following this approach, the result of projecting $\mathcal{P}_{\mathrm{v}}(\boldsymbol{k},\tau) = \langle \hat{\mathrm{v}}(\boldsymbol{k},\tau)\hat{\mathrm{v}}(\boldsymbol{k},\tau)\rangle$ is

$$\begin{aligned}
\langle \hat{\mathrm{v}}(\boldsymbol{k},\tau)\hat{\mathrm{v}}(\boldsymbol{k},\tau)\rangle_{\mathrm{div}} &= \int \mathrm{d}^n\mathbf{x}\,\mathrm{d}^n\mathbf{y}\, e^{-\mathrm{i}\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{y})}\ \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle_{\mathrm{reg}} \\
&= \int \mathrm{d}^n\boldsymbol{\xi}\,\mathrm{d}^n\mathbf{r}\, e^{-\mathrm{i}\boldsymbol{k}\cdot\mathbf{r}}\ \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle_{\mathrm{reg}}\,, \\
&\left(\text{where}\quad \boldsymbol{\xi} := \frac{\mathbf{x}+\mathbf{y}}{2}\,,\ \mathbf{r} = \mathbf{x}-\mathbf{y}\,,\right) \\
&= \int_{\mathbb{R}^n} \mathrm{d}^n\boldsymbol{\xi} \int_{\mathbb{R}^n} \mathrm{d}^n\mathbf{r}\, e^{-\mathrm{i}\boldsymbol{k}\cdot\mathbf{r}}\ \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle_{\mathrm{reg}}\,, \\
&\approx (\Omega_n)^2 \int_0^\infty \mathrm{d}\xi\ \xi^{n-1} \int_0^\infty \mathrm{d}r\, r^{n-1}\, e^{-\mathrm{i}kr}\ \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle_{\mathrm{reg}}\,.
\end{aligned} \tag{3.19}$$

The first integral in (3.19) is divergent and can be identified as an infinite volume from the center of mass frame. To solve it, dimensional regularization needs to be applied. Thus, by using the approximation (3.6) we can define the volume integral

$$V_{\mathrm{div}}(\nu,l) := \int_0^l \mathrm{d}\xi\ \xi^{2\nu-1}\,, \tag{3.20}$$

and, as $l\to\infty$, the volume turns out divergent. This infinite volume can be regularized via the μ function of Ref. [15],

$$\mu(s) := \int_1^\infty \mathrm{d}z\ z^{-s} \overset{\text{(a.c.)}}{=} \frac{1}{s-1}\,, \quad s\in\mathbb{C}\setminus\{1\}\,. \tag{3.21}$$

First, we perform a change of variable in terms of $z = \sqrt{\xi^2+1}$; $z\,\mathrm{d}z = \xi\,\mathrm{d}\xi$, to write

$$\begin{aligned}
V_{\mathrm{div}}(\nu,l) &= \int_1^{z(l)} \mathrm{d}z\ z\left(z^2-1\right)^{\nu-1}\,, \quad z(l) = \sqrt{l^2+1}\,, \\
&\approx \int_1^{z(l)} \mathrm{d}z\ z^{2\nu-1}\,,
\end{aligned}$$

By taking $l\to\infty$, then $z(l)\to\infty$, so the integral can be written as

$$\begin{aligned}
\lim_{l\to\infty} V_{\mathrm{div}}(\nu,l) &= \int_1^\infty \mathrm{d}z\ z^{2\nu-1}\,, \\
&\overset{\text{(a.c.)}}{=} \frac{1}{-2\nu}\,.
\end{aligned} \tag{3.22}$$

Therefore, we can define the regularized volume, via the analytic continuation (3.22), as

$$V_{\mathrm{reg}}(\nu) := -\frac{1}{2\nu}\,. \tag{3.23}$$

The second integral in (3.19) can be identified with the position correlator, as well as with a *Green function*, both in the relative position frame (rpf). This quantity can in principle be divergent. Let it be defined as

$$G^{(\mathrm{rpf})}_{\mathrm{div}}(\nu) := \int_0^\infty \mathrm{d}r\, r^{n-1}\, e^{-\mathrm{i}kr}\ \langle \hat{\mathrm{v}}(x)\hat{\mathrm{v}}(y)\rangle_{\mathrm{reg}}(\nu)\,. \tag{3.24}$$

By expanding the sum of the expression (3.14) and using the approximations (3.15), (3.6), when $n\to 3$, it can be found that

$$
\begin{aligned}
G_{\text{div}}^{(\text{rpf})}(\nu) \approx & \frac{A^{(1)}}{\mathrm{i}^{\alpha(1)}} \int_0^{\infty} \mathrm{d}r \; r^{n+\alpha(1)-1} e^{-\mathrm{i}kr} \\
& - \frac{A^{(1)}}{2\,\mathrm{i}^{\alpha(1)}} \int_0^{\infty} \mathrm{d}r \, r^{n+\alpha(1)+1} e^{-\mathrm{i}kr} \left[\frac{1}{(r+\omega)^2} + \frac{1}{(r-\omega)^2} \right], \\
\approx & \frac{A^{(1)}}{\mathrm{i}^{\alpha(1)+s} k^s} \int_0^{\infty} \mathrm{d}(\mathrm{i}kr) \; (\mathrm{i}kr)^{s-1} e^{-(\mathrm{i}kr)} \\
& - \frac{A^{(1)}}{2\,\mathrm{i}^{\alpha(1)}} \int_0^{\infty} \mathrm{d}r \; r^2 e^{-\mathrm{i}kr} \left[\frac{1}{(r+\omega)^2} + \frac{1}{(r-\omega)^2} \right], \qquad (3.25) \\
s := & \; n + \alpha(1) \,.
\end{aligned}
$$

Also, in order to obtain this expression (3.25), the following criteria has been applied:

(i) In the first integral, one can see that $\alpha(1) = -2n - (n+1)\varepsilon + 2\delta + 4 \approx -2$, such that the exponent $n + \alpha(1) - 1 \approx 0$, when $n \; \rightarrow 3$, at order zero in slow-roll. In spite of that, we take $\alpha(1) = -2 - 4\varepsilon + 2\delta$ at order one in slow-roll, on the grounds that in this fashion the integral relates the space dimension $n = 3$ with the slow-roll parameters. Thus, in this way the integral captures the physics of the quasi-de Sitter space-time [11].

(ii) However, in the second integral the approximation $\alpha(1) \approx -2$ at order zero in slow-roll is taken, since in this manner the pole can be tamed more easily. In addition, on account of the existence of the pole at $r = \omega$, the estimation that this pole carries the significant part of the divergent contribution is a rational approximation since the first integral already carries the non-divergent contribution.

In view of this criteria, the first integral in (3.25) can be calculated by a Gamma function. On the other hand, the second integral requires a more cautious treatment because of the regular pole at $r = \omega$. To treat this pole an integration along a quarter of a circle of radius R must be performed in the first quadrant of the complex plane, the path also needs to avoid the pole at $r = \omega$ by a small semi-circle of radius ϵ. This contour is shown at figure 1. The integrand is defined as

$$
g(z) := e^{-\mathrm{i}kz} \, z^2 \left[\frac{1}{(z+\omega)^2} + \frac{1}{(z-\omega)^2} \right]. \qquad (3.26)
$$

The calculation is derived in the appendix C.

By inserting the results (A 2)-(A 3) into (A 1), and then arranging the imaginary and real parts, the integral of interest is found to be

$$
\int_0^{\infty} \mathrm{d}r \, g(r) = \pi \, k\omega^2 \cos(k\omega) + 2\pi \, \omega \sin(k\omega) \,. \qquad (3.27)
$$

This expression can be replaced into (3.25), along with the Gamma function solution previously mentioned, to define the regularized Green function in the relative position frame as

$$
\begin{aligned}
G_{\text{reg}}^{(\text{rpf})}(\boldsymbol{k}; \tau) := & A^{(1)} \left[\frac{\Gamma(s)}{\mathrm{i}^{\alpha(1)+s} \, k^s} - \frac{1}{\mathrm{i}^{\alpha(1)}} \left(\frac{\pi}{2} k\omega^2 \cos(k\omega) + \pi\omega \sin(k\omega) \right) \right], \qquad (3.28) \\
s = & \; 4 - 2\nu \,.
\end{aligned}
$$

[11] The relevance of this parameters stems from the fact that the quantized field ν arises from a perturbation in quasi-de Sitter space, such that the slow-roll parameters measure the deviation from quasi-de Sitter to the de Sitter background.

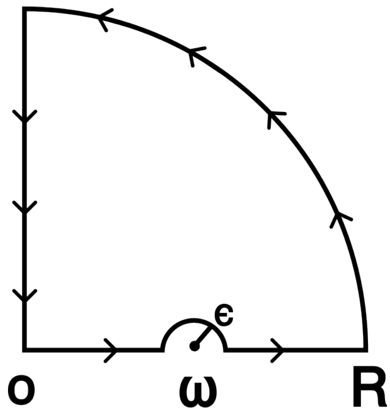

Figure 1. Contour integration in the first quadrant of the complex plane.

Returning to the equation (3.19), the results (3.23) and (3.28) can be applied to write the regularized result as

$$\begin{aligned}\mathcal{P}_{\mathrm{v}}^{(\mathrm{reg})}(\boldsymbol{k},\tau) &:= (\Omega_n)^2\, V_{\mathrm{reg}}(\nu)\, G_{\mathrm{reg}}^{(\mathrm{rpf})}(\boldsymbol{k};\tau)\,,\\ &= \frac{1}{2\nu}\,\frac{(\nu)_2}{(\frac{2\nu+\mu-2}{2})_2\,(\frac{2\nu+\mu-1}{2})_2}\,\frac{\Gamma(2\nu+\mu)}{\Gamma(2\nu+1)}\,(-\tau)^{2\nu-3}\\ &\quad\left[\frac{\Gamma(-2\nu+4)}{\mathrm{i}\,k^{-2\nu+4}}-\frac{\pi}{2}k\,(-2\tau)^2\cos(-2k\tau)-\pi(-2\tau)\sin(-2k\tau)\right]\,. \end{aligned} \tag{3.29}$$

Even though this result is particular to the case where $2\nu \approx 3$, the regularized power spectrum explicitly shows the relationship between the dimension of space n and the slow-roll parameters via the parameter ν (2.15).

In addition, (3.29) can be interpreted as a result that extends the validity of the known power spectrum of the MS variable, $\mathcal{P}_{\mathrm{v}} = |\mathrm{v}_{\boldsymbol{k}}|^2$, to all possible scales, $k \in [0, \infty]$. Meanwhile, the usual primordial power spectrum (2.21) is only valid in the super-Hubble limit when the scale of perturbations is bigger than the Hubble radius —when, the momenta satisfies $\frac{k}{aH} \ll 1$. Furthermore, the regularized correlators in position space (3.16) and in momentum space (3.29) can be employed to explore scenarios of fundamental nature, in contrast to the purely phenomenological approach of slow-roll inflation. For instance, the cosine and sine functions in (3.29) show an oscillatory behavior when $\tau \to -\infty$, which modifies the *bare* nature of the usual power spectrum

$$\mathcal{P}_{\mathrm{v}}(\boldsymbol{k},\tau) \to \frac{1}{2k}\,, \quad \tau \to -\infty\,, \tag{3.30}$$

with a Bunch-Davies vacuum prescription. This *bare* spectrum shows that at large scales ($k \to 0$) the correlation between points in the space-time increases but at small scales ($k \to \infty$) points are weakly correlated. Thus, with this vacuum prescription the fluctuations were scale invariant for large scales in the far past, when perturbations where deeply into the horizon.

On the contrary, the oscillatory nature of the regularized power spectrum (3.29) in the far past is enhanced by the factors $(-2\tau)^2$ and (-2τ), as

$$\mathcal{P}_{\mathrm{v}}^{(\mathrm{reg})}(\boldsymbol{k},\tau) \approx \frac{2}{3}\left(\frac{1}{\mathrm{i}\,k}-\frac{\pi}{2}k\,(-2\tau)^2\cos(-2k\tau)-\pi(-2\tau)\sin(-2k\tau)\right). \tag{3.31}$$

In consequence, in the far past the vacuum fluctuations were strongly correlated in an oscillatory scale-variant manner that grew with the scale factor encoded in τ. This can be interpreted as a *back-reaction* between the non-empty vacuum of the field and the space-time, which is commonly found when trying to regularize the stress-energy tensor in semi-classical gravity (Ref. [12]).

Additionally, the physical effects of a back-reaction are more noticeable when the Bunch-Davies prescription is violated.

(b) Spectral index

As mentioned in the introduction an important result of inflation is the quasi-invariant power spectrum measured by the spectral index (2.25). Hence, the regularized power spectrum should be able to address this phenomenological behavior. Indeed, this is the case.

To derive this phenomenological result the equation (3.29) is approximated in the de Sitter limit, $-\tau \approx (aH)^{-1}$ and to order zero in slow-roll —by means of the relation (3.6), $2\nu \approx 3$— to obtain

$$\mathcal{P}_{\mathrm{v}}^{(\mathrm{reg})}(\boldsymbol{k},\tau) \approx \frac{2}{3}\left[\frac{1}{\mathrm{i}\,k} - \frac{2\pi k}{(aH)^2}\cos\left(\frac{2k}{aH}\right) - \frac{2\pi}{(aH)}\sin\left(\frac{2k}{aH}\right)\right]. \tag{3.32}$$

Using (3.32) we can evaluate the regularized [12] *normalized power spectrum* (2.22) at horizon crossing $k = aH$, as

$${}^{(\mathrm{reg})}\Delta^2_{\mathcal{R}} := \frac{k^3}{2\pi^2}\mathcal{P}_{\mathrm{v}}^{(\mathrm{reg})}(\boldsymbol{k},\tau)\,\mathrm{z}^{-2}\bigg|_{k=aH}, \tag{3.33}$$

$$\approx c_0\,\frac{H^2}{\varepsilon}\bigg|_{k=aH}, \tag{3.34}$$

where c_0 is a constant[13]. A straightforward calculation of the spectral index of (3.34) shows ${}^{(\mathrm{reg})}n_{\mathrm{s}} = 1 - 2\varepsilon - \eta$.

Therefore, the spectral index for the regularized result is the same as for the usual case (2.25) once the de Sitter limit at zero order in slow-roll is taken. One key difference is that this regularized result did not depend on the super-Hubble limit approximation to yield the desired quasi-invariant behavior.

4. Conclusion

The main result of this research consists of obtaining a regularized power spectrum that successfully reproduces the expected phenomenological quasi-invariant behavior through the techniques of QFT in a dynamical background. This work represents a significant improvement upon the foundational exact-solution paradigm established in these pages by Bunch and Davies [7], advancing the field at the mathematical, phenomenological, and fundamental levels.

At the mathematical level, we have moved beyond the local point-splitting expansion utilized in the pioneering 1978 derivation. By performing an exact Fourier transform and regularizing divergent quantities through the separation of polynomial divergences in the center-of-mass frame, we provide a global treatment of the correlators. This method utilizes the principle of meromorphic continuation—rather than local coincidental subtraction—to ensure that finite results are an intrinsic consequence of the wave equation's analytic structure.

At the phenomenological level, this work demonstrates that the quasi-invariant behavior of the power spectrum is more robust to the inflationary scenario than previously thought. Because our results did not depend on the specific Bunch-Davies vacuum condition nor the super-Hubble approximation for the calculation of the correlator itself, we conclude that the observed power spectrum is largely independent of the initial vacuum state at the level of two-point observables. Of course, the super-Hubble approximation remains essential for the preservation of comoving

[12] Of course, the definition (3.33) implicitly assumes $\mathcal{P}_{\mathcal{R}}^{(\mathrm{reg})}(\boldsymbol{k},\tau) := \mathcal{P}_{\mathrm{v}}^{(\mathrm{reg})}(\boldsymbol{k},\tau)\,\mathrm{z}^{-2}$.

[13] The value of the constant amplitude of the regularized spectrum is complex, given by
$c_0 := -\frac{1}{6\pi^2}(\mathrm{i} + 2\pi\,\cos(2) + 2\pi\,\sin(2))$.
Nevertheless, it does not carry physical significance as the real amplitude of the primordial spectrum $\mathcal{A}_{\mathrm{s}}$ can be regarded as a free parameter that is actually obtained from observation (Ref. [16]). This is the so-called renormalization condition of Ref. [6].

perturbations outside the horizon, but their initial regularization is now shown to be a universal feature of the exact mode solutions.

At the fundamental level, this research proposes a new paradigm for the regularization of divergent quantities in curved space-time. By bypassing the traditional reliance on the minimal subtraction of radiative corrections—which often introduces arbitrary renormalization scales—we establish a non-perturbative path toward finite results. This represents progress in the direction of semi-classical gravity, suggesting that the "paradigm of regularization" introduced here can be extended to other dynamical backgrounds where standard perturbative methods fail to maintain predictability.

A brief comment on the variational formalism provided in section 2 can be made regarding its possible connection to the gauge-invariant approach of perturbation theory in GR [8,10], as well as with the approach of perturbative ADM [8,9]. It is of particular interest to describe a general framework where one can *translate* between the mathematical languages of these various approaches, potentially reconciling the exact-solution paradigm of the 1970s with modern computational frameworks for the Theory of Cosmological Perturbations.

Finally, future lines of work include: computing regularized corrections to the power spectrum at one-loop to further test the stability of the meromorphic extension, applying the regularization techniques introduced here to other scenarios of semi-classical gravity (such as black hole evaporation), and investigating the physical consequences of the back-reaction for non-Bunch-Davies vacua.

A. First Variation

From (1.3) the lagrangian density of General Relativity, as a functional that depends on the metric and the material field(s), can be decoupled as

$$\mathcal{L}_{\mathrm{GR}} := \mathcal{L}_{\mathrm{g}} + \mathcal{L}_{\mathrm{m}}\,, \tag{A 1a}$$

$$\mathcal{L}_{\mathrm{g}}\left(g^{\alpha\beta}\right) := \sqrt{-g}\,\kappa R\Big|_{g^{\alpha\beta}(x)}\,, \tag{A 1b}$$

$$\mathcal{L}_{\mathrm{m}}\left(g^{\alpha\beta}\right) := \sqrt{-g}\,L_{\mathrm{m}}(\phi)\Big|_{g^{\alpha\beta}(x)}\,. \tag{A 1c}$$

Consequently, is clear that a variation in the metric such as (1.5) corresponds to a functional variation in this quantities. Thus, we can define the Einstein (Ein) tensor and the Energy-Momentum (EM) tensor as functional derivatives of their respective densities (A 1),

$$\begin{aligned} G_{\mu\nu}\Big|_{g^{\alpha\beta}} &:= \frac{1}{\kappa\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\mathcal{L}_{\mathrm{g}}\left(g^{\alpha\beta}\right), \\ &= \left(R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R\right)\Big|_{g^{\alpha\beta}}\,, \end{aligned} \tag{A 2a}$$

$$T_{\mu\nu}\Big|_{g^{\alpha\beta}} := -\frac{2}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\mathcal{L}_{\mathrm{m}}\left(g^{\alpha\beta}\right). \tag{A 2b}$$

The variation of the action $\mathcal{S}_{\mathrm{GR}}$ to first order results in the tensor

$$\begin{aligned} E_{\mu\nu}\left(g^{\alpha\beta}\right) &:= \frac{2}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\mathcal{L}_{\mathrm{EH}}\left(g^{\alpha\beta}\right), \\ &= 2\kappa\, G_{\mu\nu} - T_{\mu\nu}\Big|_{g^{\alpha\beta}}\,, \end{aligned} \tag{A 3}$$

defined as the *first background tensor*. This tensor vanishes when evaluating at $g^{\alpha\beta}(p)$, for every point in the background space-time ($p \in \mathcal{M}_0$), according to the variational principle $\delta\mathcal{S}_{\mathrm{GR}} = 0$.

B. In-in formalism

In the In-in or Schwinger-Keldysh formalism, the correlators of an observable $W(t)$ in the interaction picture are calculated as

$$\langle W(t_\star)\rangle_0 = \langle W_{\rm int}(t_\star)\rangle\,, \tag{A 1a}$$

$$\langle W(t_\star)\rangle_1 = -2\,\mathrm{Im}\left\{\int_{t_0^-}^{t_\star} \mathrm{d}t_1\ \langle H_{\rm int}(t_1) W_{\rm int}(t_\star)\rangle\right\}, \tag{A 1b}$$

$$\begin{aligned}\langle W(t_\star)\rangle_2 &= \int_{t_0^-}^{t_\star} \mathrm{d}t_2 \int_{t_0^+}^{t_\star} \mathrm{d}t_1\ \langle H_{\rm int}(t_2)\, W_{\rm int}(t_\star)\, H_{\rm int}(t_1)\rangle \\ &\quad - 2\,\mathrm{Re}\left\{\int_{t_0^-}^{t_\star} \mathrm{d}t_2 \int_{t_0^-}^{t_\star} \mathrm{d}t_1\ \langle H_{\rm int}(t_1)\, H_{\rm int}(t_2)\, W_{\rm int}(t_\star)\rangle\right\},\end{aligned} \tag{A 1c}$$

for the non-interacting, tree-level and one-loop interactions, respectively. In the previous calculations (2.31)-(2.32), the term (A 1c) for the one-loop interaction was the one represented.

C. Contour integral

The integral of the function $g(z)$ in equation (3.26) along the whole contour is zero because there is no residue inside it. Furthermore, the contour $\mathcal{C}$ may be divided into the following curves: (a) a real line from 0 to $\omega - \epsilon$; (b) a semi-circle of radius ϵ centered at ω; (c) another real line from $\omega + \epsilon$ to R; (d) a quarter of a circle of radius R centered at the origin; and (e) an imaginary line connecting iR to the origin.

From this, the integral can be written as

$$\oint_{\mathcal{C}} \mathrm{d}z\, g(z) = \int_0^{\omega-\epsilon} \mathrm{d}x\, g(x) + \int_{\mathcal{C}_\epsilon} \mathrm{d}z\, g(z) + \int_{\omega+\epsilon}^{R} \mathrm{d}x\, g(x) + \int_{\mathcal{C}_R} \mathrm{d}z\, g(z) + \int_R^0 \mathrm{d}(\mathrm{i}y)\, g(\mathrm{i}y)\,. \tag{A 1}$$

Then, by taking the limit when $R \to \infty$ and $\epsilon \to 0$, the integrals tend to

$$\int_0^{\omega-\epsilon} \mathrm{d}x\, g(x) + \int_{\omega+\epsilon}^{R} \mathrm{d}x\, g(x) \Longrightarrow \int_0^{\infty} \mathrm{d}x\, g(x)\,, \tag{A 2a}$$

$$\int_{\mathcal{C}_\epsilon} \mathrm{d}z\, g(z) \Longrightarrow \frac{1}{\mathrm{i}} \int_0^{\pi} \mathrm{d}\theta \lim_{\mathsf{z}\to 0} \Big[\mathsf{z}\, g(\mathsf{z}+\omega)\Big]\,, \quad \mathsf{z} := \epsilon\, e^{\mathrm{i}\theta}\,, \tag{A 2b}$$

$$\int_{\mathcal{C}_R} \mathrm{d}z\, g(z) \Longrightarrow \mathrm{i} \int_0^{\pi/2} \mathrm{d}\theta \lim_{\mathsf{Z}\to \infty} \Big[\mathsf{Z}\, g(\mathsf{Z})\Big]\,, \quad \mathsf{Z} := R\, e^{\mathrm{i}\theta}\,, \tag{A 2c}$$

$$\int_R^0 g(\mathrm{i}y)\, \mathrm{d}(\mathrm{i}y) \Longrightarrow -\mathrm{i} \int_0^{\infty} \mathrm{d}y\, g(\mathrm{i}y)\,. \tag{A 2d}$$

A straightforward calculation gives the limits inside the integrals as

$$\lim_{\mathsf{z}\to 0} \Big[\mathsf{z}\, g(\mathsf{z}+\omega)\Big] = e^{-\mathrm{i}k\omega}\left(-\mathrm{i}k\omega^2 + 2\omega\right), \tag{A 3a}$$

$$\lim_{\mathsf{Z}\to \infty} \Big[\mathsf{Z}\, g(\mathsf{Z})\Big] = 0\,. \tag{A 3b}$$

Author Contributions

The author is the sole contributor to this work, responsible for the conceptualization, mathematical derivations, development of the regularization methodology, and the writing and editing of the manuscript.

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